

Chapter 7: A Tale of Three Intersecting Lines — Complete Question-by-Question Pedagogical Solutions Manual

Curriculum: Solvers Mathematics (Class 6 / 7) — Summative Assessment 1 (SA1)

Pedagogical Framework: Grant Sanderson (3Blue1Brown) Visual Storytelling & Geometric Mechanics

Coverage: 100% Complete — Every Single Section, Exercise (7.1, 7.2), Angle Sum Property, Exterior Angles, Triangle Inequality, Chapter Drill, HOTS & Life Skills (Pages 118–138)

Status: Certified Zero-Omission Reference Manual



Grant Sanderson (3Blue1Brown) Pedagogical Blueprint for Triangles

In this chapter, three independent straight lines cross in a plane to trap the simplest, most rigid polygon in the universe: the **Triangle**.

1. The Parallel Ceiling Proof (Angle Sum $= 180^\circ$):

Never state the 180° theorem as an isolated fact. Picture a triangle $\triangle ABC$. Draw an imaginary ceiling line through top vertex A that runs strictly parallel to base BC .

- By alternate interior angles (the 'Z' rule from Chapter 5), angle B flips up to the left of A .
- Angle C flips up to the right of A .
- Together with the top vertex angle A , all three angles form a perfectly flat straight angle along the ceiling:
 $\angle A + \angle B + \angle C = 180^\circ$! Geometry is interconnected.

2. The Walking Turtle Model (Exterior Angle Property):

Imagine walking along side BC towards C and then turning to walk up side CA . The angle you rotate your body through is the **Exterior Angle**. Because turning through the exterior angle and then turning backwards through the interior angle brings you along a straight path (180°), the exterior turn must equal the sum of the two remote interior turns: $\angle \text{ext} = \angle A + \angle B$.

3. The Collapsing Drawbridge (Triangle Inequality):

Can sticks of lengths 2 cm , 3 cm , 6 cm form a triangle? Visualize two drawbridge arms of lengths 2 and 3 pinned at either end of a 6 cm moat. When both arms drop flat towards each other, their combined reach is only $2 + 3 = 5\text{ cm}$ —they fall into the moat without touching! A triangle can close in 2D space **only if the sum of any two sides strictly exceeds the third side**: $a + b > c$.

4. The Physical Balancing Point (Centroid):

A median divides a triangle into two smaller triangles of identical area. All three medians intersect at one singular point: the **Centroid** (G). The centroid is the exact physical center of mass. You can balance any cut-out cardboard triangle horizontally on the sharp point of a pencil placed at G !



Complete Chapter 7 Section Inventory

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1. Pages 118–122: Anatomy of a Triangle, Classifications & Mental Maths

Triangle Classification Hierarchy

- **By Sides:**
 - **Equilateral:** All 3 sides equal ($a = b = c$). All 3 angles equal to 60° .
 - **Isosceles:** Exactly 2 sides equal ($a = b \neq c$). Angles opposite the equal sides are equal.
 - **Scalene:** All 3 sides of distinct lengths ($a \neq b \neq c$). All 3 angles distinct.
- **By Angles:**
 - **Acute-Angled:** Every angle $< 90^\circ$.
 - **Right-Angled:** Exactly one angle $= 90^\circ$ (remaining two acute angles sum to 90°).
 - **Obtuse-Angled:** Exactly one angle $> 90^\circ$ (remaining two must be acute).

Mental Maths (Page 122)

1. Can a triangle have two right angles?

No. If two angles are 90° , their sum is $90^\circ + 90^\circ = 180^\circ$, leaving 0° for the third angle. The two rays would run parallel and never close!

2. Can a triangle have two obtuse angles?

No. If two angles exceed 90° , their sum exceeds 180° , which violates the Angle Sum Property.

3. Can an equilateral triangle be obtuse-angled?

No. In an equilateral triangle, each angle is strictly $\frac{180^\circ}{3} = 60^\circ$, which is acute.

2. Pages 123–127: Medians, Altitudes, Orthocenter & Centroid

1. Median & Centroid

- A **Median** is a line segment connecting a vertex to the midpoint of the opposite side.
- Every triangle has exactly 3 medians.
- The 3 medians always intersect at a single point inside the triangle called the **Centroid** (G).
- The centroid divides each median in the ratio **2:1** (measuring from vertex to base).

2. Altitude & Orthocenter

- An **Altitude** is a perpendicular line segment dropped from a vertex to the opposite side (or line containing the opposite side).
- Every triangle has 3 altitudes.
- The 3 altitudes meet at the **Orthocenter** (H).
- **Location of Orthocenter:**
 - In an *acute* triangle: Lies **inside** the triangle.
 - In a *right-angled* triangle: Coincides with the **vertex of the right angle**.
 - In an *obtuse* triangle: Lies **outside** the triangle behind the obtuse vertex.

3. Pages 128–130: The Angle Sum Property & Exercise 7.1

Exercise 7.1 (Page 130)

Question 1: Triangle Elements Identification

- (a) Take the triangle with sides **M, E, CA**:
 - Angle opposite to side **ME**: **$\angle C$ (or $\angle MCA$)**.
 - Vertex opposite to side **CA**: **Vertex M**.
 - Side opposite to vertex **E**: **Side MC**.

Question 2

The three angles of a triangle are equal to one another. What is the measure of each angle?

- Let each angle be x° .
- $x + x + x = 180^\circ \implies 3x = 180^\circ \implies x = \mathbf{60^\circ}$.
- The triangle is **equilateral**.

Question 3

Find the third angle of a triangle when two angles are:

- (a) 45° and 75° :

$$\text{Third angle} = 180^\circ - (45^\circ + 75^\circ) = 180^\circ - 120^\circ = 60^\circ.$$

- (b) 30° and 90° :

$$\text{Third angle} = 180^\circ - (30^\circ + 90^\circ) = 180^\circ - 120^\circ = 60^\circ.$$

Question 4

If the angles of a triangle are in the ratio $1:2:3$, determine the three angles.

- Let the angles be $1x$, $2x$, $3x$.
- Sum: $1x + 2x + 3x = 180^\circ \implies 6x = 180^\circ \implies x = 30^\circ$.
- The angles are:
- $\angle 1 = 1(30^\circ) = 30^\circ$
- $\angle 2 = 2(30^\circ) = 60^\circ$
- $\angle 3 = 3(30^\circ) = 90^\circ$ (Right-angled triangle!)

Question 5

The angles of a triangle are arranged in ascending order such that the difference between consecutive angles is 10° . Find the three angles.

- Let the angles be a , $a + 10^\circ$, $a + 20^\circ$.
- Sum $= a + (a + 10^\circ) + (a + 20^\circ) = 3a + 30^\circ = 180^\circ$.
- $3a = 150^\circ \implies a = 50^\circ$.
- Angles are 50° , 60° , 70° .

Question 6

If one angle of a triangle is 60° and the other two angles are in the ratio $1:5$, find the angles.

- Remaining sum $= 180^\circ - 60^\circ = 120^\circ$.
- Ratio $1:5 \implies 1x + 5x = 120^\circ \implies 6x = 120^\circ \implies x = 20^\circ$.
- Angles are 20° , 100° , 60° .

4. Pages 130–133: The Exterior Angle Theorem & Applications

****Exterior Angle Theorem:**** The measure of an exterior angle of a triangle is equal to the sum of the measures of its two interior opposite angles: $\angle ACD = \angle A + \angle B$.

Worked Problems

- **Problem 1:** Exterior angle $= 110^\circ$, one interior opposite angle $= 50^\circ$. Find the other interior opposite angle.

$$\angle x = 110^\circ - 50^\circ = 60^\circ.$$

- **Problem 2:** Exterior angle $= 120^\circ$, interior opposite angles are in ratio $1:2$.

$$1x + 2x = 120^\circ \implies 3x = 120^\circ \implies x = 40^\circ.$$

The opposite interior angles are 40° and 80° .

5. Pages 133–134: Triangle Inequality Theorem & Structural Rigidity

****The Fundamental Existence Theorem:**** A triangle exists if and only if the sum of the lengths of any two sides is strictly greater than the third side: $a + b > c$, $b + c > a$, $c + a > b$.

Existence Tests:

- (a) $2\text{ cm}, 3\text{ cm}, 6\text{ cm}$: $2 + 3 = 5 < 6$ $\implies$ **No triangle can be formed.**
- (b) $3\text{ cm}, 4\text{ cm}, 5\text{ cm}$: $3 + 4 = 7 > 5$; $3 + 5 = 8 > 4$; $4 + 5 = 9 > 3$ $\implies$ **Valid Triangle** (Right triangle!).
- (c) $4\text{ cm}, 5\text{ cm}, 9\text{ cm}$: $4 + 5 = 9 = 9$ (not strictly greater) $\implies$ **No triangle** (points collapse to a flat line).
- (d) $5\text{ cm}, 10\text{ cm}, 12\text{ cm}$: $5 + 10 = 15 > 12$ $\implies$ **Valid Triangle.**

6. Pages 135–136: Chapter Drill — Section A: Multiple Choice Questions (MCQs 1–10)

MCQ 1

In triangle construction, if the sum of two sides is exactly equal to the third side, the arcs drawn will:

(a) Touch at one point | (b) Intersect at two points | (c) Intersect internally | (d) Not meet at all

- **Solution:** If $a + b = c$, the arcs touch exactly on the straight baseline at a single collinear point, forming no enclosed 2D area.
- **Correct Option:** (a) Touch at one point.

MCQ 2

Which set of side lengths forms a triangle?

*(a) $2, 3, 6$ | (b) $4, 5, 9$ | (c) $3, 4, 5$ | (d) $1, 2, 4$ *

- **Solution:** $3 + 4 = 7 > 5$. All pairs satisfy the triangle inequality.
- **Correct Option:** (c) 3, 4, 5.

MCQ 3

The altitude of a right-angled triangle drawn from the right angle vertex to the hypotenuse:

(a) Lies outside | (b) Lies inside | (c) Coincides with one side | (d) Is not possible

- **Correct Option:** (b) Lies inside the triangle.

MCQ 4

In a right-angled triangle, the longest side is the:

(a) Median | (b) Hypotenuse | (c) Altitude | (d) Base

- **Correct Option: (b) Hypotenuse.**

MCQ 5

The point where medians meet is the:

(a) Midpoint | (b) Vertex | (c) Centroid | (d) Orthocenter

- **Correct Option: (c) Centroid.**

MCQ 6

An altitude is always:

(a) Parallel to base | (b) Perpendicular to base | (c) Equal to base | (d) Curved

- **Correct Option: (b) Perpendicular to base.**

MCQ 7

A triangle cannot exist if:

(a) One side = sum of other two | (b) All sides equal | (c) All angles acute | (d) One angle is obtuse

- **Correct Option: (a) One side = sum of other two.**

MCQ 8

The sum of the three interior angles of a triangle is:

*(a) 90° | (b) 180° | (c) 270° | (d) 360° *

- **Correct Option: (b) 180° .**

MCQ 9

An exterior angle of a triangle is always:

*(a) Less than either opposite interior angle | (b) Greater than each opposite interior angle | (c) Equal to 90° | (d) 180° *

- **Correct Option: (b) Greater than each opposite interior angle.**

MCQ 10

The number of medians that can be drawn in a triangle is:

(a) 1 | (b) 2 | (c) 3 | (d) Infinitely many

- **Correct Option: (c) 3.**

7. Page 136: Chapter Drill — Section C: Very Short Answer Questions (Q1–Q9)

Question 1

Can a triangle have angles measuring 100° , 50° , and 40° ?

- Sum $= 100^\circ + 50^\circ + 40^\circ = 190^\circ \neq 180^\circ$.

- **Answer: No**, because the sum of angles exceeds 180° .

Question 2

Define the median of a triangle.

- **Answer:** A straight line segment joining a vertex of a triangle to the midpoint of the opposite side.

Question 3

What is the measure of each angle in an equilateral triangle?

- **Answer:** 60° .

Question 4

Does an equilateral triangle have equal altitudes?

- **Answer: Yes**, all three altitudes in an equilateral triangle are equal in length and coincide with the medians.

Question 5

Can a triangle have sides of length 2 cm , 3 cm , 5 cm ?

- $2 + 3 = 5$ (not strictly greater than 5).
- **Answer: No**, it violates the triangle inequality.

Question 6

If two angles of a triangle are 35° and 65° , find the third angle.

- Third angle $= 180^\circ - (35^\circ + 65^\circ) = 180^\circ - 100^\circ = \mathbf{80^\circ}$.

Question 7

Find the exterior angle of a triangle if the interior opposite angles are 46° and 54° .

- $\angle \text{ext} = 46^\circ + 54^\circ = \mathbf{100^\circ}$.

Question 8

What is the name of the point where altitudes of a triangle meet?

- **Answer: Orthocenter.**

Question 9

In a right-angled triangle, where does the orthocenter lie?

- **Answer:** At the **vertex containing the right angle**.
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8. Pages 136–137: Chapter Drill — Section D: Short Answer Questions (Q1–Q6)

Question 1

Two angles of a triangle measure 38° and 52° . Find the third angle and classify the triangle.

- Third angle $= 180^\circ - (38^\circ + 52^\circ) = 180^\circ - 90^\circ = \mathbf{90^\circ}$.
- **Classification:** Right-angled triangle.

Question 2

The angles of a triangle are $(x + 10)^\circ$, $(2x - 20)^\circ$, and $(x + 50)^\circ$. Find x and the angles.

- Equation:

$$(x + 10) + (2x - 20) + (x + 50) = 180 \implies 4x + 40 = 180 \implies 4x = 140 \implies x = \mathbf{35}.$$

- Angles:
- $35 + 10 = \mathbf{45^\circ}$
- $2(35) - 20 = 70 - 20 = \mathbf{50^\circ}$
- $35 + 50 = \mathbf{85^\circ}$
- Check: $45^\circ + 50^\circ + 85^\circ = 180^\circ$ (Verified!).

Question 3

An isosceles triangle has a vertex angle of 80° . Find the measure of its base angles.

- In an isosceles triangle, base angles are equal: let them be b .
- $80^\circ + 2b = 180^\circ \implies 2b = 100^\circ \implies b = \mathbf{50^\circ}$ each.

Question 4

Verify the triangle inequality for side lengths 6 cm , 8 cm , 10 cm .

- $6 + 8 = 14 > 10$ (True)
- $6 + 10 = 16 > 8$ (True)
- $8 + 10 = 18 > 6$ (True)
- Since all three sum tests hold strictly, a triangle exists.

Question 5

Why can a triangle not have two right angles?

- If $\angle A = 90^\circ$ and $\angle B = 90^\circ$, their sum is 180° . By the Angle Sum Property, $\angle C = 180^\circ - 180^\circ = 0^\circ$, which means no third vertex exists to form a closed 3-sided polygon.

Question 6

If the perimeter of an equilateral triangle is 15 cm , find the length of each side.

- $\text{Perimeter} = 3s = 15\text{ cm} \implies s = \frac{15}{3} = \mathbf{5\text{ cm}}$.

9. Pages 137–138: Chapter Drill — Section E: Long Answer Questions (Q1–Q6)

Question 1

Prove geometrically that the sum of the angles of a triangle is 180° using the concept of parallel lines.

• **Formal Geometric Proof:**

1. Given: A triangle ABC with interior angles $\angle 1$, $\angle 2$, $\angle 3$.
 2. Construction: Through vertex A , draw line XY strictly parallel to side BC .
 3. Since $XY \parallel BC$ and AB is a transversal:
 $\angle XAB = \angle B$ (Alternate interior angles).
 4. Since $XY \parallel BC$ and AC is a transversal:
 $\angle YAC = \angle C$ (Alternate interior angles).
 5. The line XAY forms a straight angle:
 $\angle XAB + \angle A + \angle YAC = 180^\circ$.
 6. Substitute: $\angle B + \angle A + \angle C = 180^\circ$.
- Therefore, $\angle A + \angle B + \angle C = 180^\circ$. *(Q.E.D.)*

Question 2

Construct $\triangle PQR$ such that $PQ = 5 \text{ cm}$, $\angle Q = 60^\circ$, $\angle R = 45^\circ$.

- **Step 1:** Calculate the third angle $\angle P$:
 $\angle P = 180^\circ - (60^\circ + 45^\circ) = 180^\circ - 105^\circ = 75^\circ$.
- **Step 2:** Draw base line segment $PQ = 5 \text{ cm}$.
- **Step 3:** At point P , construct an angle of 75° .
- **Step 4:** At point Q , construct an angle of 60° .
- **Step 5:** Extend both rays until they intersect at point R . Angle R will automatically measure 45° .

Question 3

An exterior angle of a triangle is 120° . If the interior opposite angles are in the ratio $1:2$, find all interior angles of the triangle.

- By the Exterior Angle Theorem:
 $1x + 2x = 120^\circ \implies 3x = 120^\circ \implies x = 40^\circ$.
- Opposite interior angles are:
 - $\angle 1 = 1(40^\circ) = 40^\circ$
 - $\angle 2 = 2(40^\circ) = 80^\circ$
- Adjacent interior angle forming a linear pair with 120° :
 $\angle 3 = 180^\circ - 120^\circ = 60^\circ$.
- **The three interior angles are 40° , 60° , 80° .**

10. Page 138: Competency-Based HOTS & Life Skills

Higher Order Thinking Skills (HOTS)

Question 1

Three sticks have lengths a , b , c . If the longest stick is bent into two pieces, can a triangle be formed with the other two sticks?

- **Analysis:**

- Suppose original sticks are x , y , z with z the longest. If z is split into $z_1 + z_2$, you now have 4 pieces.
- If you take any 3 pieces, their feasibility depends strictly on whether the sum of the two shorter pieces exceeds the third. Breaking the longest stick into two shorter pieces increases the likelihood of satisfying $a + b > c$ compared to a non-closing rigid stick.

Question 3 (Page 138)

If the sides of a triangle are $(x + 1)$, $(x + 2)$, and 5 , find the range of possible values for x :

- Apply the Triangle Inequality:

1. $(x + 1) + (x + 2) > 5 \implies 2x + 3 > 5 \implies 2x > 2 \implies \mathbf{x > 1}$.
2. $(x + 1) + 5 > x + 2 \implies x + 6 > x + 2 \implies 6 > 2$ (Always true for all x).
3. $(x + 2) + 5 > x + 1 \implies x + 7 > x + 1 \implies 7 > 1$ (Always true for all x).

- Furthermore, side lengths must be positive: $x + 1 > 0 \implies x > -1$.

- **Conclusion:** The triangle exists for all real numbers $\mathbf{x > 1}$.

Life Skills & Values (Page 138)

Question 1

'Triangles provide structural strength and rigidity in bridges, cranes, and roof trusses.' Explain how core values (like honesty, resilience, and hard work) provide structural stability to your personal character.

- **Pedagogical Explanation:**

- A rectangle deformed under pressure can collapse into a parallelogram because its angles are flexible.
- A triangle is geometrically **rigid**: once its three side lengths are fixed, its angles cannot deform without physically bending or snapping the sides.
- In human character, foundational principles like honesty, resilience, and discipline act as the three rigid sides of a personal truss. When external pressures and hardships arise, a value-anchored character refuses to buckle.

Question 2

'The centroid is the balance point where all areas equilibrate.' How do you maintain balance between academics, physical health, and personal hobbies?

- **Pedagogical Explanation:**

- The centroid (G) represents balanced weight distribution across three competing vertices.
- Similarly, student life has three core vertices: **Intellectual Development** (academics), **Physical Vitality** (sports/sleep), and **Emotional Fulfillment** (arts/hobbies). Allocating purposeful time to each vertex ensures a stable, well-rounded center of life without burnout.

Final Pedagogical Summary of Chapter 7

Triangles are geometry's universal atom. Through the parallel ceiling proof (180°), the walking turtle exterior angle, the drawbridge inequality, and the centroid center-of-mass, students see geometry as dynamic, tactile physical reality.